COS 201 — Computer Programming I

Week 2: Development Environment and First Java Program

1. Folder structure

```
COS201-Programming-I/
├── README.md
├── week01/
│   └── FirstProgram.java
└── week02/
    ├── HelloWorld.java
    ├── PrintExample.java
    └── ErrorExample.java
```

2. HelloWorld.java (practical)

```
/*
 * COS 201 - Week 2 Practical: Development Environment and First Java Program
 */
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
        System.out.println("Welcome to COS 201.");
        System.out.println("I am learning Java programming.");
        System.out.println();
        System.out.println("COURSE: COS 201");
        System.out.println("COURSE TITLE: Computer Programming I");
        System.out.println("PROGRAMME: Computer Science");
        System.out.println("WEEK: 2");
        System.out.println("TOPIC: Development Environment and Java");
        System.out.println("GOAL: To learn how to write Java programs.");
    }
}
```

Output:

```
Hello, World!
Welcome to COS 201.
I am learning Java programming.

COURSE: COS 201
COURSE TITLE: Computer Programming I
PROGRAMME: Computer Science
WEEK: 2
TOPIC: Development Environment and Java
GOAL: To learn how to write Java programs.
```

3. PrintExample.java (practice exercise)

```
// Predicted output:
// COS 201
// Java
// print() stays on the same line, so "COS " and "201" join into "COS 201".
// println("201") then moves to a new line, so "Java" is printed on the next line.
public class PrintExample {
    public static void main(String[] args) {
        System.out.print("COS ");
        System.out.println("201");
        System.out.println("Java");
    }
}
```

Predicted output: `COS 201` then `Java` on the next line. `print()` stays on the same line, so "COS " and "201" join; `println("201")` then moves to a new line.

4. ErrorExample.java (find and fix the error)

The error was a missing semicolon after `System.out.println("Learning Java")`. Fixed version:

```
// The original code was missing a semicolon after the println statement.
public class ErrorExample {
    public static void main(String[] args) {
        System.out.println("Learning Java");
    }
}
```

Output: `Learning Java`

5. How to compile and run

```
cd week02
javac HelloWorld.java
java HelloWorld
```

6. README.md (Week 2 section)

```
## Week 2: Development Environment and First Java Program

### Student Information
- Name: [Your full name]
- Matriculation Number: [Your matric number]
- Programme: [Your programme]
- Course: COS 201
- Week: 2

### Practical Title
Development Environment and First Java Program

### Objective
To configure the Java development environment, write a simple Java program, compile
it, execute it and submit the source code through GitHub.

### Tools Used
- Java Development Kit (JDK)
- Visual Studio Code
- Git
- GitHub Desktop
- GitHub

### Files
- `week02/HelloWorld.java`: Java source code for the practical.
- `week02/PrintExample.java`: print() vs println() exercise.
- `week02/ErrorExample.java`: missing-semicolon debugging exercise (fixed).

### How to Run
1. Open a terminal in the `week02` directory.
2. Compile the program using `javac HelloWorld.java`.
3. Run the program using `java HelloWorld`.

### Learning Outcome
I practised creating, compiling and running a Java program and submitting source code
through GitHub.
```

7. Git steps in GitHub Desktop

1. Save the files in the week02 folder of your cloned repo, and save README.md.
2. Open the Changes tab and check the files are listed.
3. Enter the summary "Complete Week 2 Java practical" and click Commit to main.
4. Click Push origin.
5. Refresh the repo page on GitHub and confirm week02/HelloWorld.java and README.md are visible.

8. Submission message

```
COS 201 – Week 2 Practical Submission

Dear Lecturer,

I am submitting my Week 2 practical assignment for COS 201 – Computer Programming I.

Student Name: [Your Full Name]
Matriculation Number: [Your Matric Number]
Programme: [Your Programme]
Assignment: Development Environment and First Java Program
GitHub Repository: [Paste your actual repository URL here]

I have included my Java source code and README documentation in the repository.
I have also compiled and executed the program to verify the output.

Thank you.
```